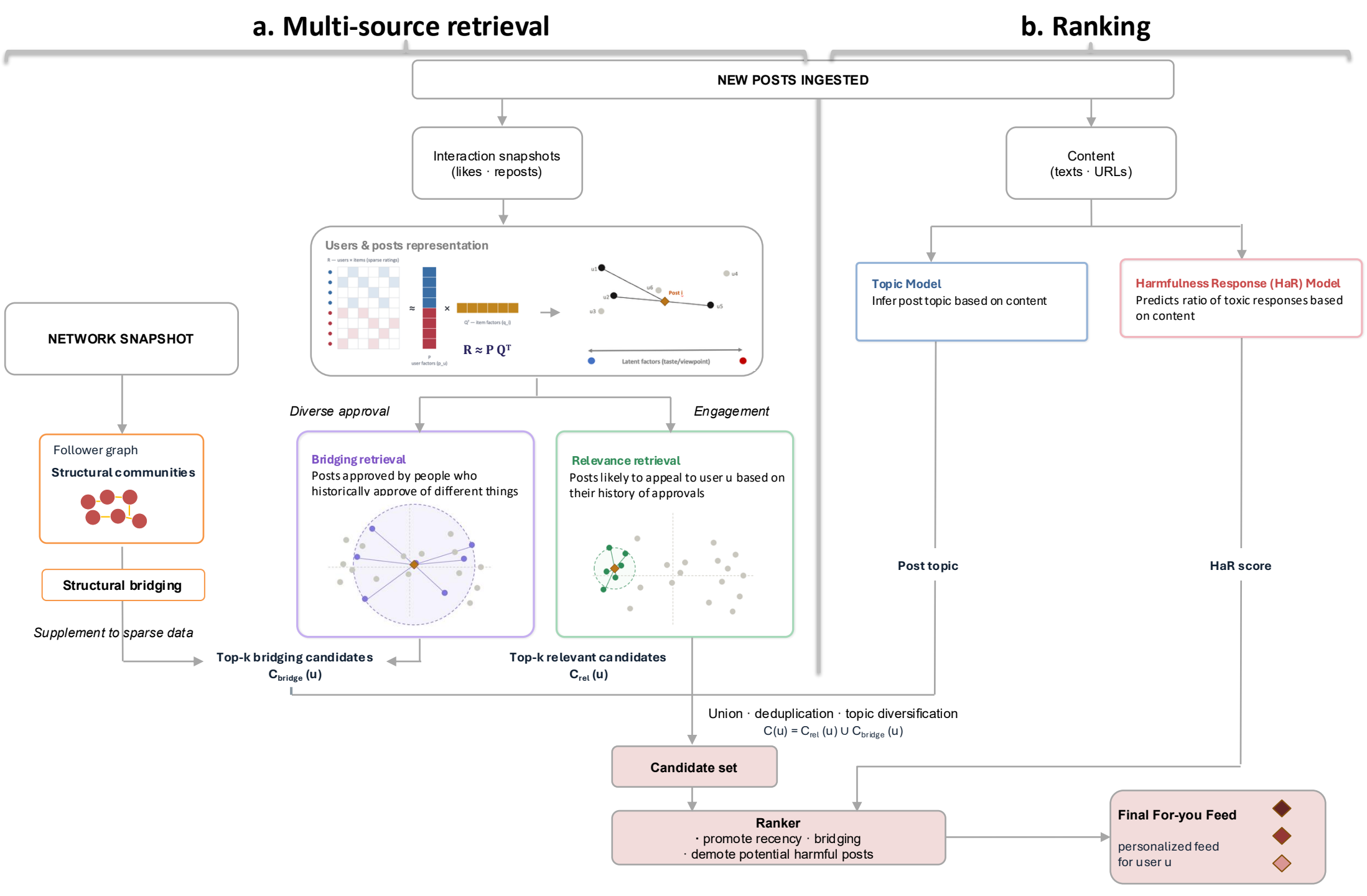
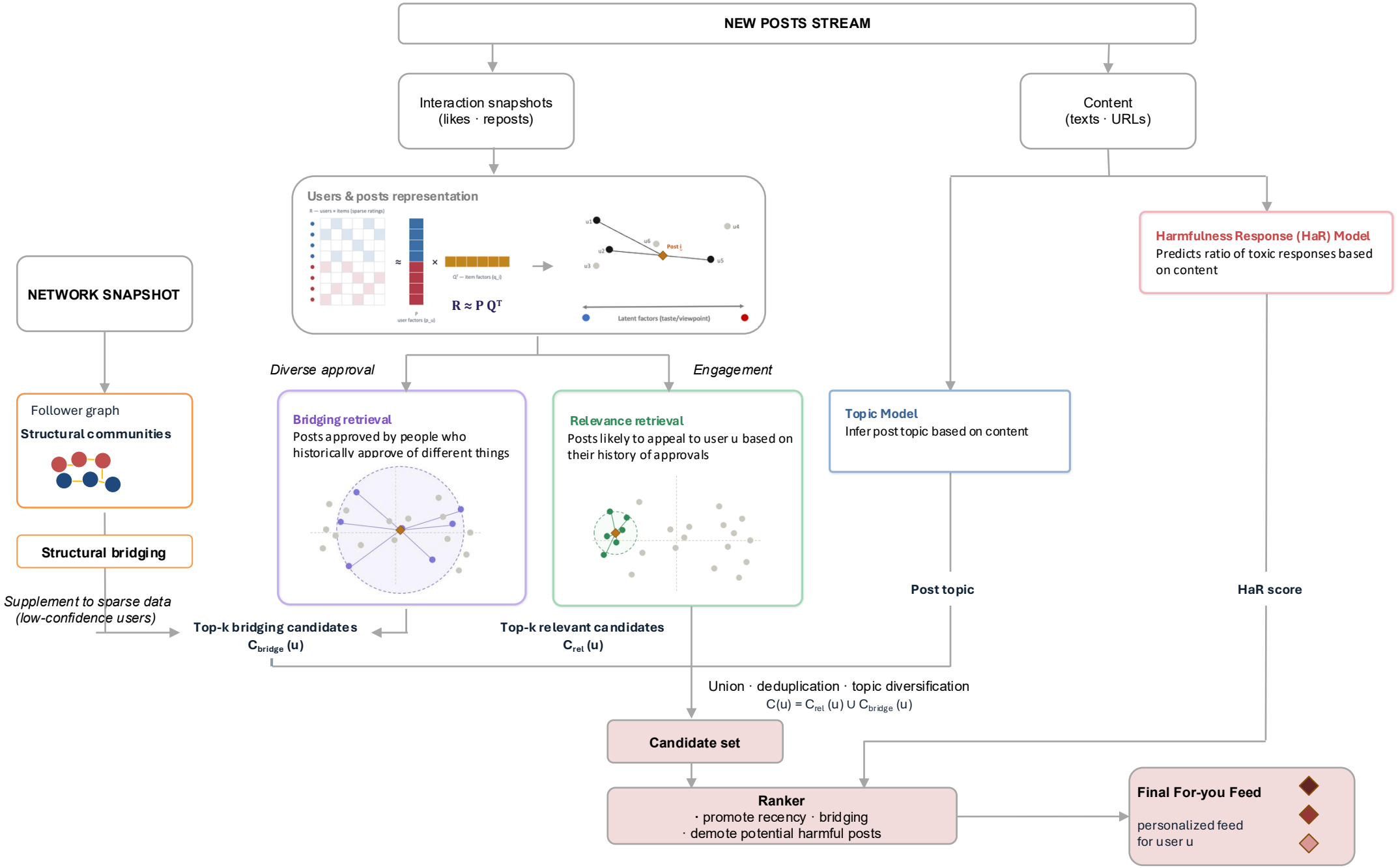
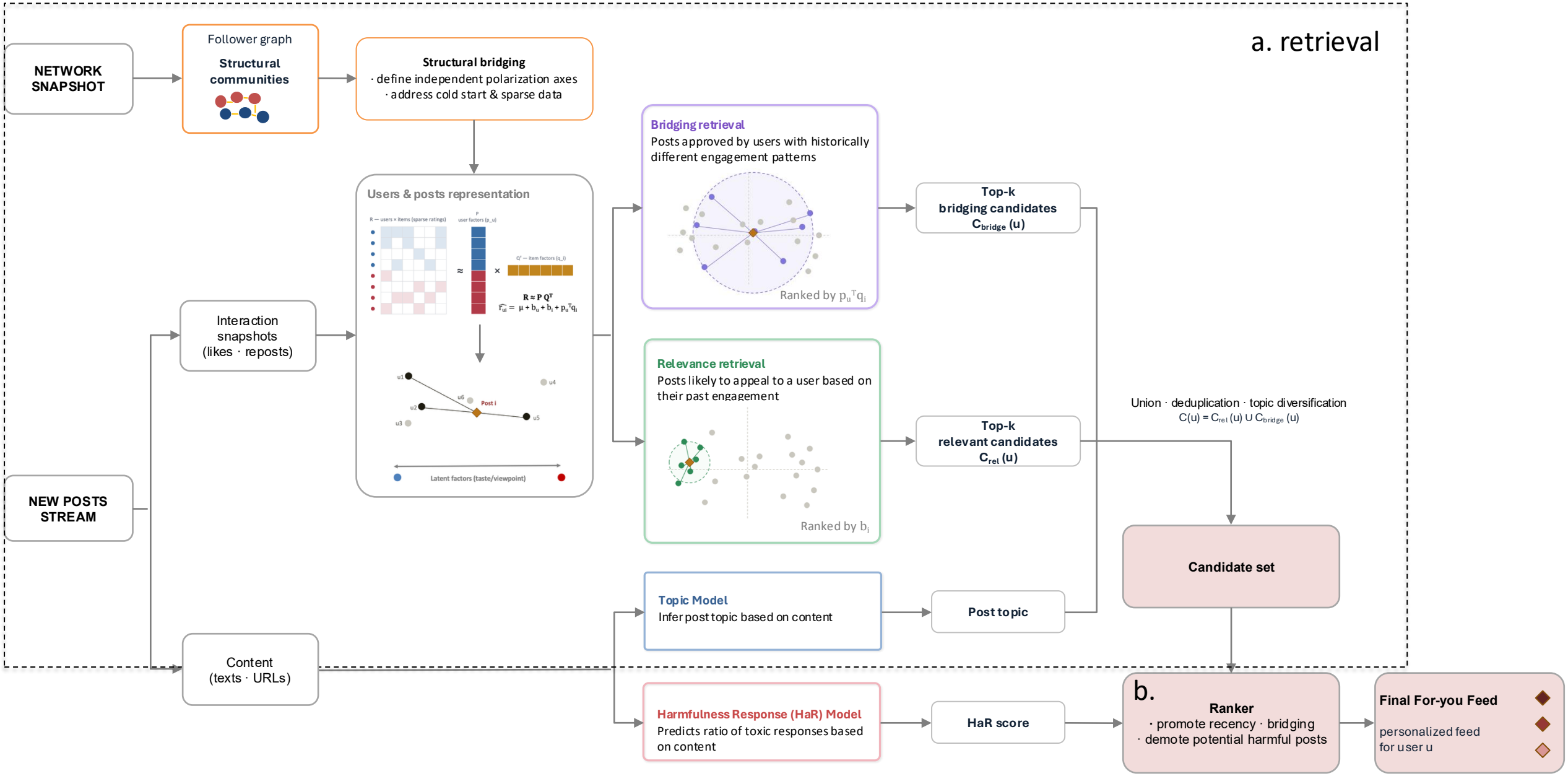
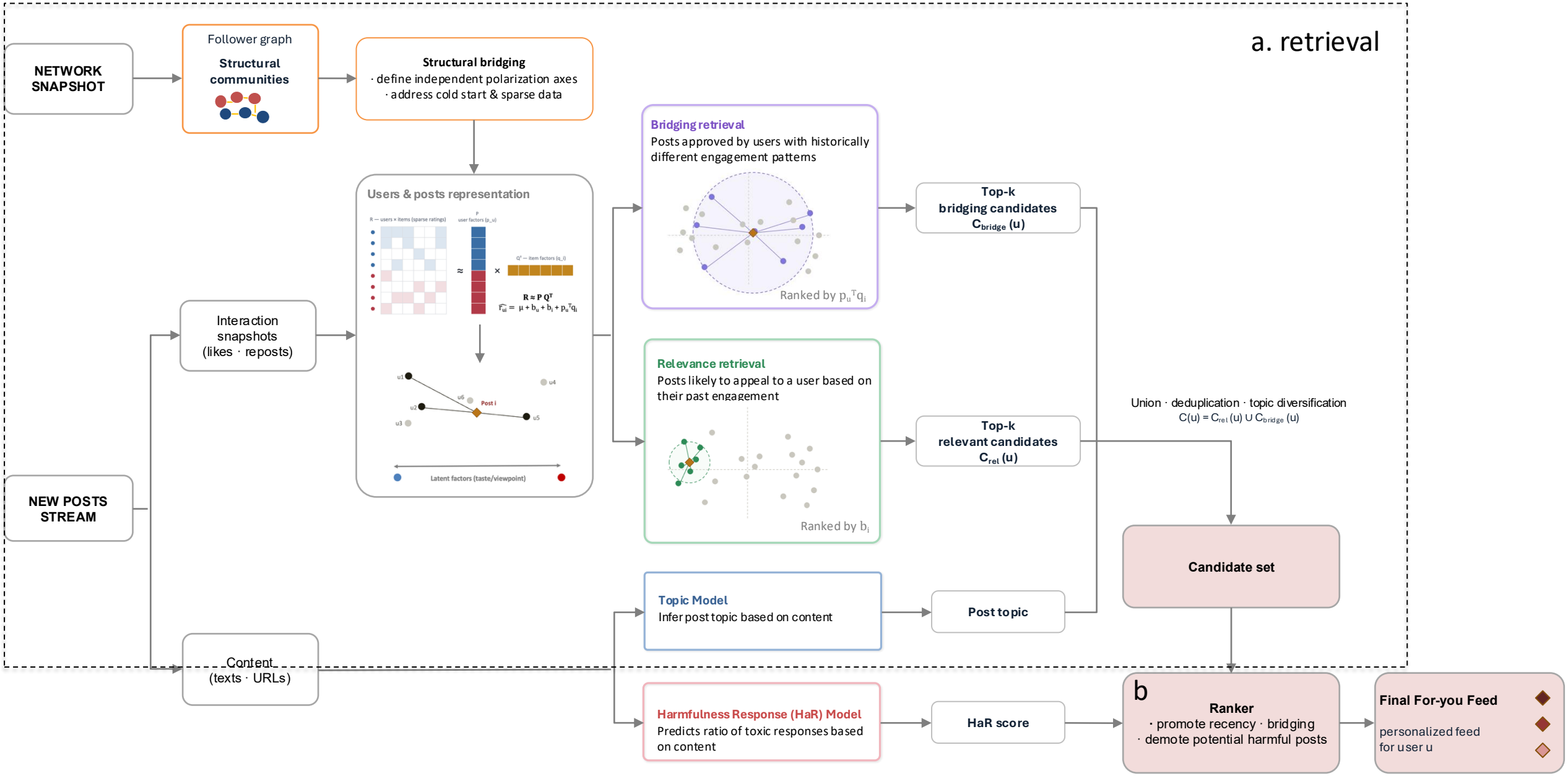

a. Multi-source retrieval
b. Ranking
NEW POSTS INGESTED
Interaction snapshots
(likes · reposts)
Content
(texts · URLs)
Users & posts representation
R ≈ P Q^T
Latent factors (taste/viewpoint)
NETWORK SNAPSHOT
Follower graph
Structural communities
Structural bridging
Supplement to sparse data
Topic Model
Infer post topic based on content
Harmfulness Response (HaR) Model
Predicts ratio of toxic responses based on content
Diverse approval
Engagement
Bridging retrieval
Posts approved by people who historically approve of different things
Relevance retrieval
Posts likely to appeal to user u based on their history of approvals
Post topic
HaR score
Top-k bridging candidates
C_bridge (u)
Top-k relevant candidates
C_rel (u)
Union · deduplication · topic diversification
C(u) = C_rel (u) ∪ C_bridge (u)
Candidate set
Ranker
· promote recency · bridging
· demote potential harmful posts
Final For-you Feed
personalized feed
for user u

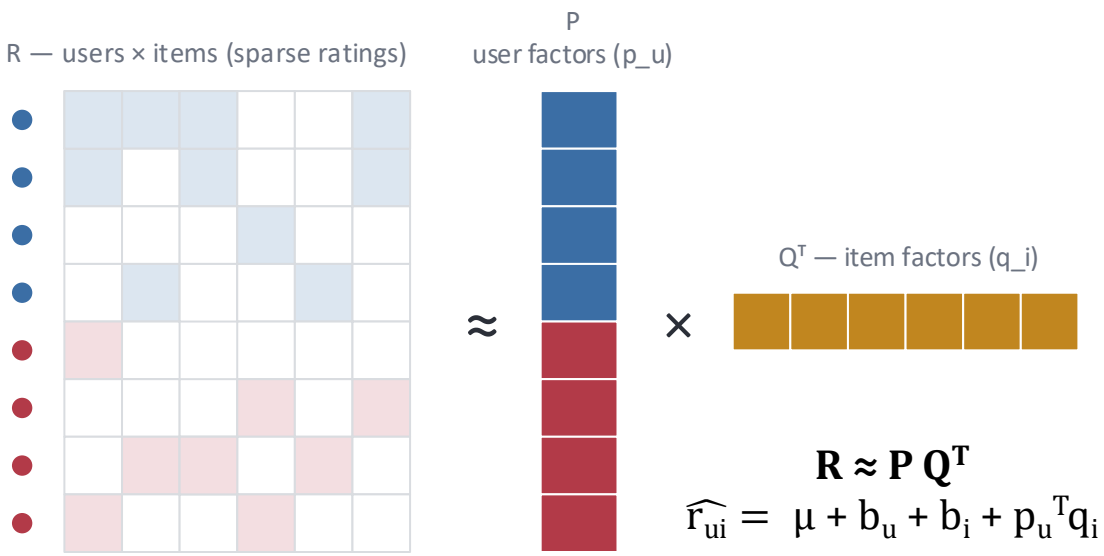

NEW POSTS STREAM
Interaction snapshots
(likes · reposts)
Content
(texts · URLs)
Users & posts representation
R ≈ P Q^T
Latent factors (taste/viewpoint)
Harmfulness Response (HaR) Model
Predicts ratio of toxic responses based on content
NETWORK SNAPSHOT
Follower graph
Structural communities
Structural bridging
Supplement to sparse data
(low-confidence users)
Diverse approval
Engagement
Bridging retrieval
Posts approved by people who historically approve of different things
Relevance retrieval
Posts likely to appeal to user u based on their history of approvals
Topic Model
Infer post topic based on content
Post topic
HaR score
Top-k bridging candidates
C_bridge (u)
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C(u) = C_rel (u) ∪ C_bridge (u)
Candidate set
Ranker
· promote recency · bridging
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Final For-you Feed
personalized feed
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a. retrieval
NETWORK SNAPSHOT
Follower graph
Structural communities
Structural bridging
· define independent polarization axes
· address cold start & sparse data
Users & posts representation
Interaction snapshots (likes · reposts)
NEW POSTS STREAM
Content (texts · URLs)
Bridging retrieval
Posts approved by users with historically different engagement patterns
Ranked by $p_u^T q_i$
Top-k bridging candidates C_{bridge} (u)
Relevance retrieval
Posts likely to appeal to a user based on their past engagement
Ranked by b_i
Top-k relevant candidates C_{rel} (u)
Union · deduplication · topic diversification
$C(u) = C_{rel}(u) \cup C_{bridge}(u)$
Candidate set
Topic Model
Infer post topic based on content
Post topic
Harmfulness Response (HaR) Model
Predicts ratio of toxic responses based on content
HaR score
b.
Ranker
· promote recency · bridging
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a. retrieval
NETWORK SNAPSHOT
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Top-k bridging candidates $C_{bridge}(u)$
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Ranked by b_i
Top-k relevant candidates $C_{rel}(u)$
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The Shared Starting Point

Both systems fit the same kind of model to a ratings matrix



$$\mathbf{R \approx P\,Q^T}$$

$$\widehat{r_{ui}} = \mu + b_u + b_i + p_u{}^T q_i$$

Shown here at rank 1 — a single latent factor per user / item, e.g. "which side of an issue do you lean toward."

predicted rating / helpfulness:

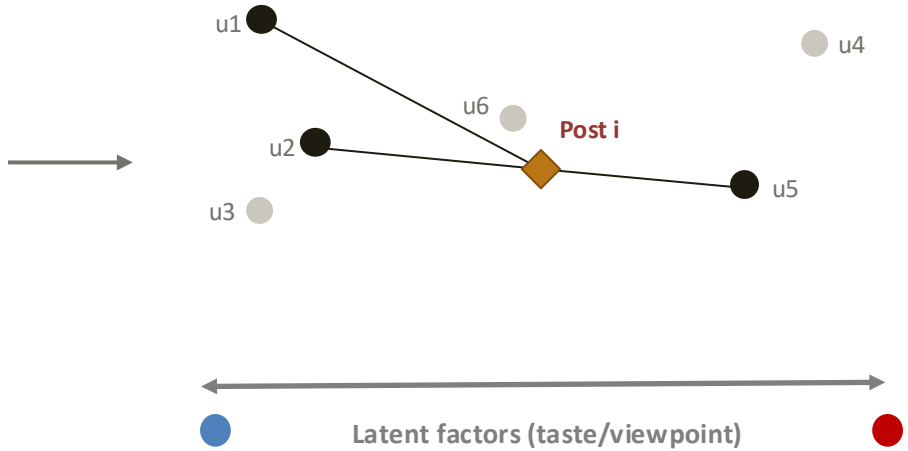
$$\text{score}(u, i) = \mu + b_u + b_i + (p_u \cdot q_i)$$

Legend — used the same way on every slide

- Group A — one cluster of users / tastes
- Group B — a different cluster of users
- Item or note vector being ranked
- Term the ranking decision is based on
- Term the model fits, but ranking ignores

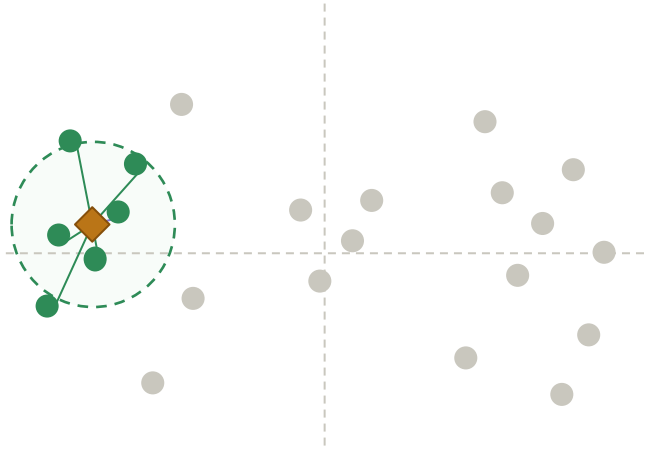
Both systems fit user vectors, item vectors, and bias terms to explain past ratings. What differs is which piece of this equation each one uses to decide what to show — that's the whole story on the next two slides.

	i1	i2	i3	i4	i5	i6
u1	5	4	5	1	2	3
u2	4	5	4	2	3	2
u3	5	5	2	1	2	3
u4	1	2	3	5	2	4
u5	2	1	5	4	3	3
u6	3	3	2	3	5	3



Factional item

one camp approves



Diverse approval

both camps approve

